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either neither

Abstract

This paper proposes adding two usability concepts to more concisely express (not) matching a list of types.

Tony Table

Before	Proposed
<pre>template<typename T> requires(same_as<T, char> or same_as<T, short>) void func(T) { ... }</pre>	<pre>template<either<char, short> T> void func(T) { ... } // or just: void func(either<char, short> auto) { ... }</pre>
<pre>template<typename T> requires(not same_as<T, short> and not same_as<T, int>) void func(T) { ... }</pre>	<pre>template<neither<short, int> T> void func(T) { ... } // or just: void func(neither<short, int> auto) { ... }</pre>

Revisions

R0: Initial version

Motivation

We've encountered the need to express a template type parameter matching one of multiple types across several projects. Whilst we can't provide sustained data on how widespread this pattern already is, we want to point to both the documentation for `same_as` on [cppreference](#) as well as [stackoverflow](#) for prior usage/requests to express this semantics.

Design Space

Given these facilities are simple wrappers around foldings of `same_as`, the only design decisions pertain to naming. We've considered the following alternative names and discarded them for the given reasons.

- `oneof/noneof` ... only partially correct as one could assume that `oneof` expresses matching exactly one given type.
- `anyof/noneof` ... express the basic idea, but are unusual spellings to circumvent collisions with the respective algorithms.

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The best names we could come up with after excluding the above have been either/neither, which admittedly look funky² when used directly (either<T, U...>) but look fine when using the “terse notation” (template<either<U...> T>, void f(either<U...> auto)).

Impact on the Standard

This proposal is a pure library addition.

Proposed Wording

Wording is relative to [N5001]. Additions are presented like **this**, removals like ~~this~~ and drafting notes like **this**.

[version.syn]

```
#define __cpp_lib_concepts 202207L-YYMM-LL // freestanding, also in <concepts>, <compare>
```

[DRAFTING NOTE: Adjust the placeholder value as needed to denote the proposal's date of adoption.]

[concepts]

Header <concepts> synopsis

[concepts.syn]

```
// all freestanding
namespace std {
    // [concepts.lang], language-related concepts
    // [concept.same], concept same_as
    template<class T, class U>
        concept same_as = see below;

    template<class T, class... U>
        concept either = (same_as<T, U> || ...);

    template<class T, class... U>
        concept neither = !either<T, U...>;

    // [concept.derived], concept derived_from
}
```

Acknowledgements

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² Not more funky than already existing concepts (e.g. derived_from<Derived, Base>) though...